



Systems + Thinking

An Overview



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Preface

This book is an overview to the area of systems thinking and theory that is designed to be accessible to a broad group of people. The book is focused upon two primary achievements; firstly providing you with the key concepts that will enable you to see the world in a whole new way from the systems perspective, what we call systems thinking. Secondly the aim is to provide you with the standardized language of systems theory through which you will be able to describe and model systems of all kind in a more coherent fashion whilst also being able to effectively communicate this to others.

This book requires no prior specific knowledge of mathematical modeling or science, as we will be starting with the very basic model of a system and then building upon this to create more sophisticated representation. The course is broken down into four main areas.

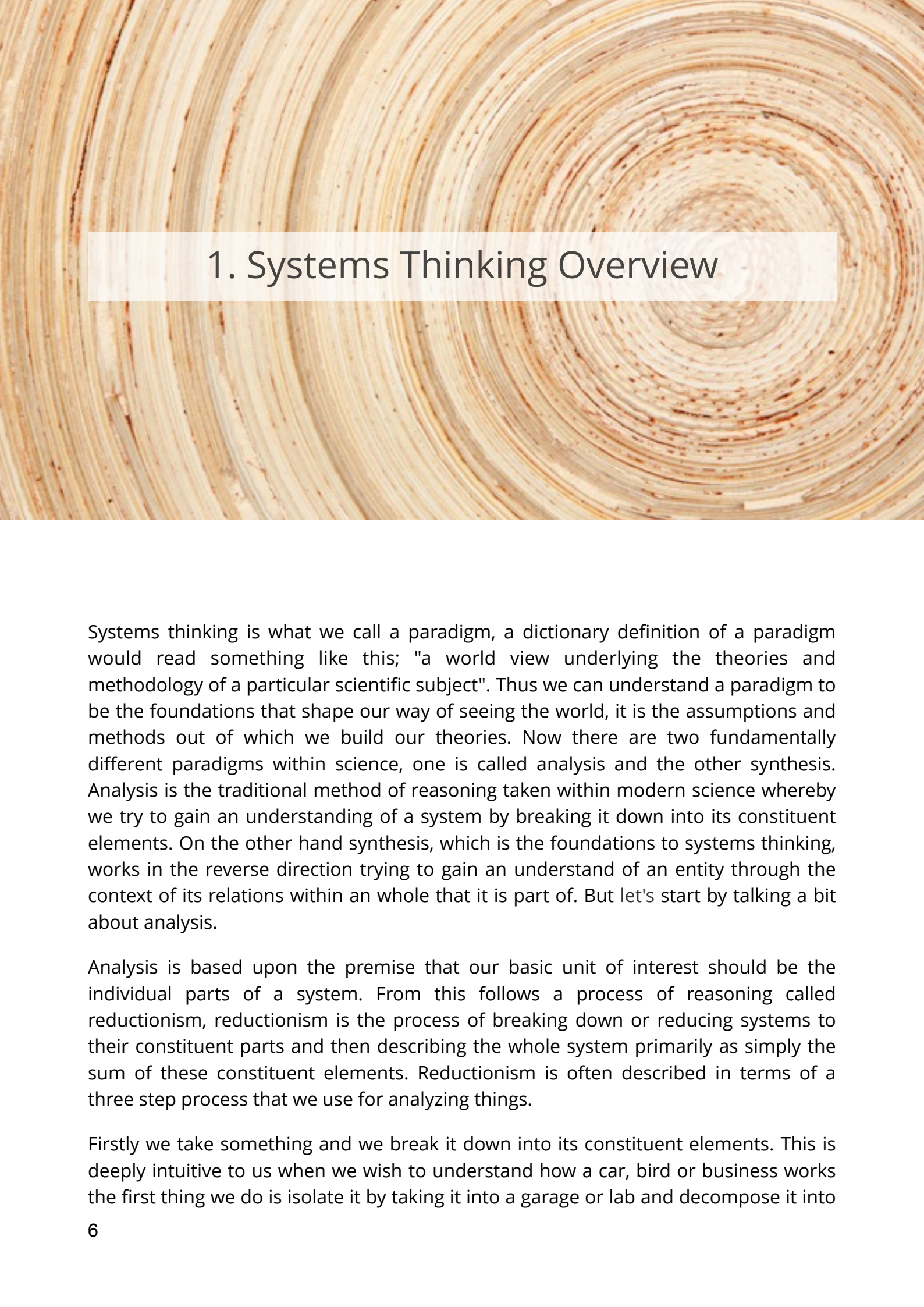
Firstly we will start the course with an overview to systems thinking making a clear distinction between our traditional methods of analytical reasoning and the alternative method of synthesis that forms the foundations of system thinking.

Next we will delve into systems theory to start building our model of a system, clearly defining what exactly a system is and is not. During the rest of this section we will build upon this model by adding the concepts of efficiency, functionality and talking about energy and entropy.

In the third section to the course we will develop our model into a more powerful framework by adding the concept of the system's environment, discussing systems boundaries, synergistic interactions and the emergence of hierarchical structure out of these synergies.

In the last section we will look at different models for capturing how systems change over time what is called system dynamics, here we will explore the ideas of feedback loops, causal loop diagrams and the phenomena of homeostasis. Finally we wrap-up the course with a discussion of systems science, looking at how and why it is of relevance to us.

By the end of this book you should have gained a whole new perspective on the world call systems thinking and will have gained an understanding of the formal language of systems theory that can be used within a wide variety of applications from engineering to business management to IT to many areas of science. Lets get started!

The background of the slide is a close-up, high-resolution image of a wood grain. The grain is composed of numerous concentric, slightly wavy rings of varying shades of light brown, tan, and beige. The texture is organic and natural, with some darker, reddish-brown spots scattered throughout the lighter wood. The lighting is even, highlighting the fine details of the wood's structure.

1. Systems Thinking Overview

Systems thinking is what we call a paradigm, a dictionary definition of a paradigm would read something like this; "a world view underlying the theories and methodology of a particular scientific subject". Thus we can understand a paradigm to be the foundations that shape our way of seeing the world, it is the assumptions and methods out of which we build our theories. Now there are two fundamentally different paradigms within science, one is called analysis and the other synthesis. Analysis is the traditional method of reasoning taken within modern science whereby we try to gain an understanding of a system by breaking it down into its constituent elements. On the other hand synthesis, which is the foundations to systems thinking, works in the reverse direction trying to gain an understand of an entity through the context of its relations within an whole that it is part of. But let's start by talking a bit about analysis.

Analysis is based upon the premise that our basic unit of interest should be the individual parts of a system. From this follows a process of reasoning called reductionism, reductionism is the process of breaking down or reducing systems to their constituent parts and then describing the whole system primarily as simply the sum of these constituent elements. Reductionism is often described in terms of a three step process that we use for analyzing things.

Firstly we take something and we break it down into its constituent elements. This is deeply intuitive to us when we wish to understand how a car, bird or business works the first thing we do is isolate it by taking it into a garage or lab and decompose it into

its constituent parts. Secondly, once we have broken the system down into its most elementary components we analyze these individual components in isolation in order to describe their properties and their functioning in isolation. Lastly we recombine these components into the original system that can now be described in terms of the properties of its individual elements.

The reductionist approach is the fundamental method behind modern science and by extension our modern understanding of the world and it has proven highly successful in many ways from understanding atoms and DNA to designing the modern corporation and nation state, but as successful as it has been it also has inherent limitations to it. Because we understand systems by breaking the parts down and isolating them, the reductionist paradigm systematically and inherently de-promotes the relationships between these components. Within this paradigm of reductionism the whole system is implicitly thought to be nothing more than the sum of its parts.

Thus analysis works well when there is a low level of interconnectivity and interdependencies within the system we are modeling, although this may be true for some systems it is certainly not always the case, many of the systems we are interested in describing have a high level of interconnectivity and interdependency, examples being ecosystems, computer networks and many types of social systems. These systems in contrast are primarily defined by the relations within the system and not the static properties of their elements, we can and often do continue to use analysis to try and describe them but the reductionist approach is not designed for this and thus we need to change our basic paradigm to one that is more focused upon these relations as opposed to the components and this is where systems and systems thinking comes in.



"Systems thinking means the ability to see the synergy of the whole rather than just the separate elements of a system and to learn to reinforce or change whole system patterns."

- Richard L. Daft

Syntheses means the combination of components or elements to form a connected whole: It is a process of reasoning that describes an entity through the context of its relations and functioning within the whole system that it is a part of. Systems thinking is this process of reasoning called syntheses and it is also referred to as being what is called holistic. Meaning that it is characterized by the belief that the parts of something are intimately interconnected and explicable only by reference to the whole.

Thus syntheses focuses on the relations between the elements, that is to say the way those elements are put together or arrange into a functioning entirety and like with analyses we can also identify a few key stages in this process of reasoning. The first step in the process is to identify the system that our object of interest is a part of, examples of this might be a bird being part of a broader ecosystem or a person being part of a greater culture. Next we try to gain a broad outline of how this whole system functions. So for example a hard drive is part of a computer and to properly understand it we need to have some understanding of the whole computer. Lastly we try to understand how the parts are interconnected and arranged to function as an entirety.

By completing this process we can identify the complex of relations within which our entity is embedded, its place and function within the whole and within systems thinking this context is considered the primary frame of reference for describing something.



2. Synthesis & Analysis

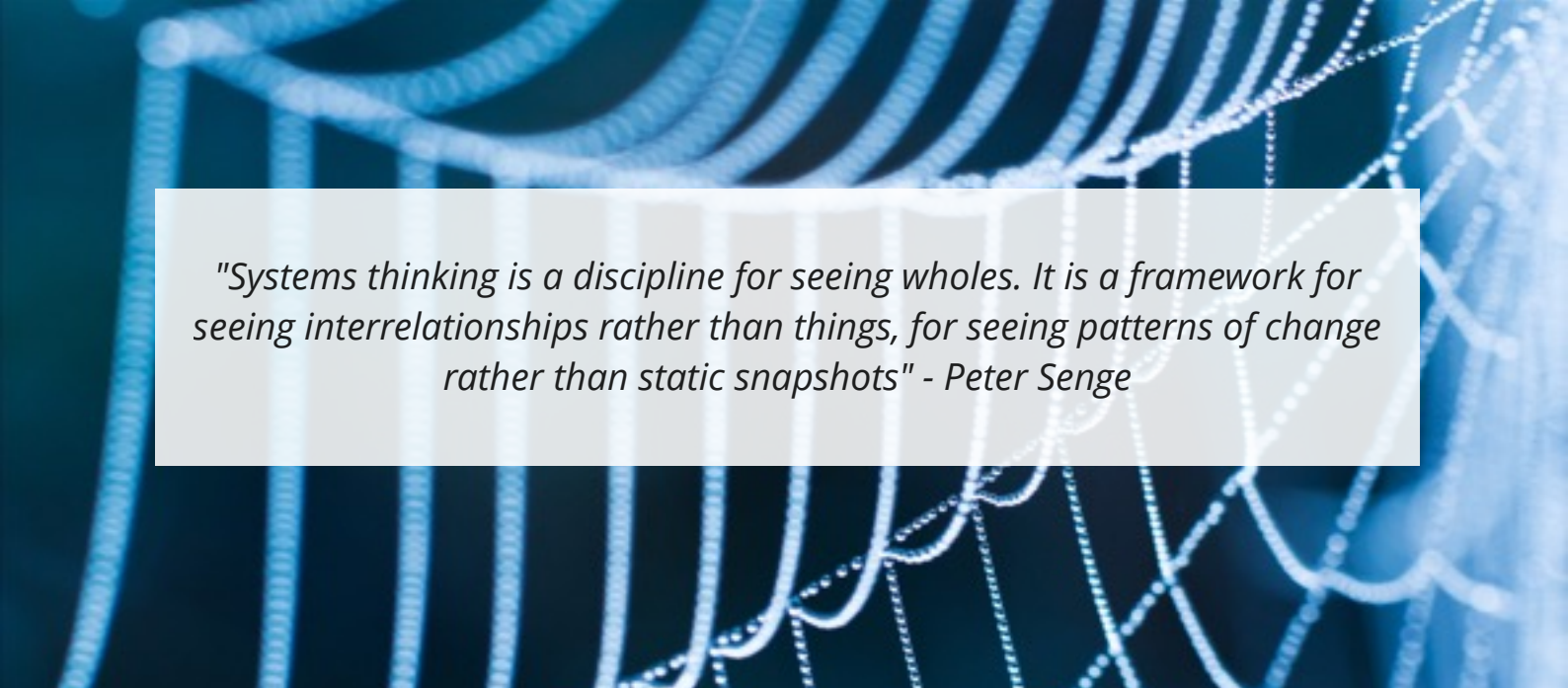
In this section we are going to continue our discussion on analysis and synthesis, digging a bit deeper into the distinction between the two. The first thing to note is that the methods of synthesis and analysis are not mutually exclusive, they should both be a part of any well developed model but each will have particular relevance depending upon the type or properties of system we are dealing with. Thus it should not be of surprise to us that physics is the home of the reductionist approach where they are often dealing with inert, static and decomposable systems, whereas ecologists that deal with highly interconnected and dynamic systems are much more inclined to systems thinking.

So some of the primary question we will be asking to determine the type of system we are dealing with and thus the appropriate method of reasoning will be firstly; Is it primary a component-based system or does it serve some common function that integrate the various elements?; Is it isolated or connected?; Is it a linear deterministic system or a non-linear non-determinate system?; And is it static or dynamic? We will be covering many of these topics in more depth later on in the book so we will just be touching on them for the moment.

Firstly are we dealing with an actual system or simply as a set of things? When we wish to talk about a composite entity, that is to say a group of things, we can describe it as either a set of objects or a system, the difference here being that a set is a group of objects that share no common function, thus we call a group of cups on a table a set of cups as they exist independently from each other. In contrast if we take the human body, again it is a composite entity, but this time the elements have been designed to

serve some common function and thus we can call it a system and we need to use systems thinking to properly understand it.

Secondly how interconnected is the system? Analysis starts from a component based view of the world and builds a description based upon the properties of these components. Synthesis in contrary focuses upon the relationships between parts thus from a systems thinking perspective we are often interested in connectivity i.e. answering the question what is connected to what and thus is best suited to systems with a high-level of interconnectivity.



"Systems thinking is a discipline for seeing wholes. It is a framework for seeing interrelationships rather than things, for seeing patterns of change rather than static snapshots" - Peter Senge

Thirdly are we dealing with a linear system or are there feedback loops? Analytical thinking searches for direct linear relations between the cause of an event and the effect. Thus we call this linear thinking. Systems thinking is more inclined to see events at the product of a complex of interacting parts where relations are often cyclical with feedback loops.

Is the system primarily static or dynamic? Analytical methods often describe entities in terms of static structures with limited reference to their development within time. Systems thinking takes a more dynamic view of things often contextualizing entities in terms of the evolutionary forces that have shaped them and thus seeing the process of development as an important phenomena with which to understand the world.

Lastly are we dealing with a system on the micro level or the macro level? Analysis breaks things down into parts and thus analytical thinking typically focuses upon analyzing and optimizing subsystems, in a believe that we can improve the whole

system by simply optimizing all of its subcomponents.

If we are dealing with a system on the macro level, what we sometime call the global level we need to use systems thinking to get a vision on of the whole system and an understanding of how the parts interrelate to achieve global functionality.



3. Sets Vs. Systems

Up until now we have been using the term system quite loosely and people often use it as a kind of catch all phrase, but in this section we are going to get a bit more rigorous to clearly define what exactly a system is and what it is not. There are many definitions for a system out there so let's take a few quick examples: Wikipedia tells us that; "A system is a set of interacting or interdependent components forming an integrated whole". Or according to the Oxford dictionary a system is; "A set of things working together as parts of a mechanism or an interconnecting network." So firstly a system is a group of parts that is to say it is a composite entity being composed of a number of things, in the language of systems theory we call these parts, elements. Next these parts are interconnected and interdependent in some way, that is to say there is a set of relationships between the elements. Lastly; that through these relations the elements are arranged in a particular way in order to perform some collective function that defines the system as a whole.

So let's take an example of this, a business organization is a type of system, it has a number of parts or elements that are the business departments, such as production, R&D, sales, accounting and so on. All of these departments are interconnected, they exchange information, resources, personnel etc. and through this exchange they are organized to perform some collective function, that is producing some good or services. There are of course many examples of systems from transportation systems to agricultural systems and health care systems. Not everything is a system though if we have a group of things that are not interconnected and do not work together then this is not a system, it is what we call a set, a simple set of elements.

An example of a set might be a pile of bricks, or a group of people waiting at a bus stop. These compositions have not been design to work together thus we describe them by simply describing the properties of each element in the set and that tells us everything we need to know, there is nothing more to the set than the simple sum of its elements. This very important feature to sets helps to make dealing with them relatively simple. Now say we took this pile of bricks and we built a house out of them, we would no longer describe them as a simple set of bricks because by building our house we have now added a set of relations, a particular arrangement to them that allows them to function as an interdependent entirety and this entirety of the house is the system.

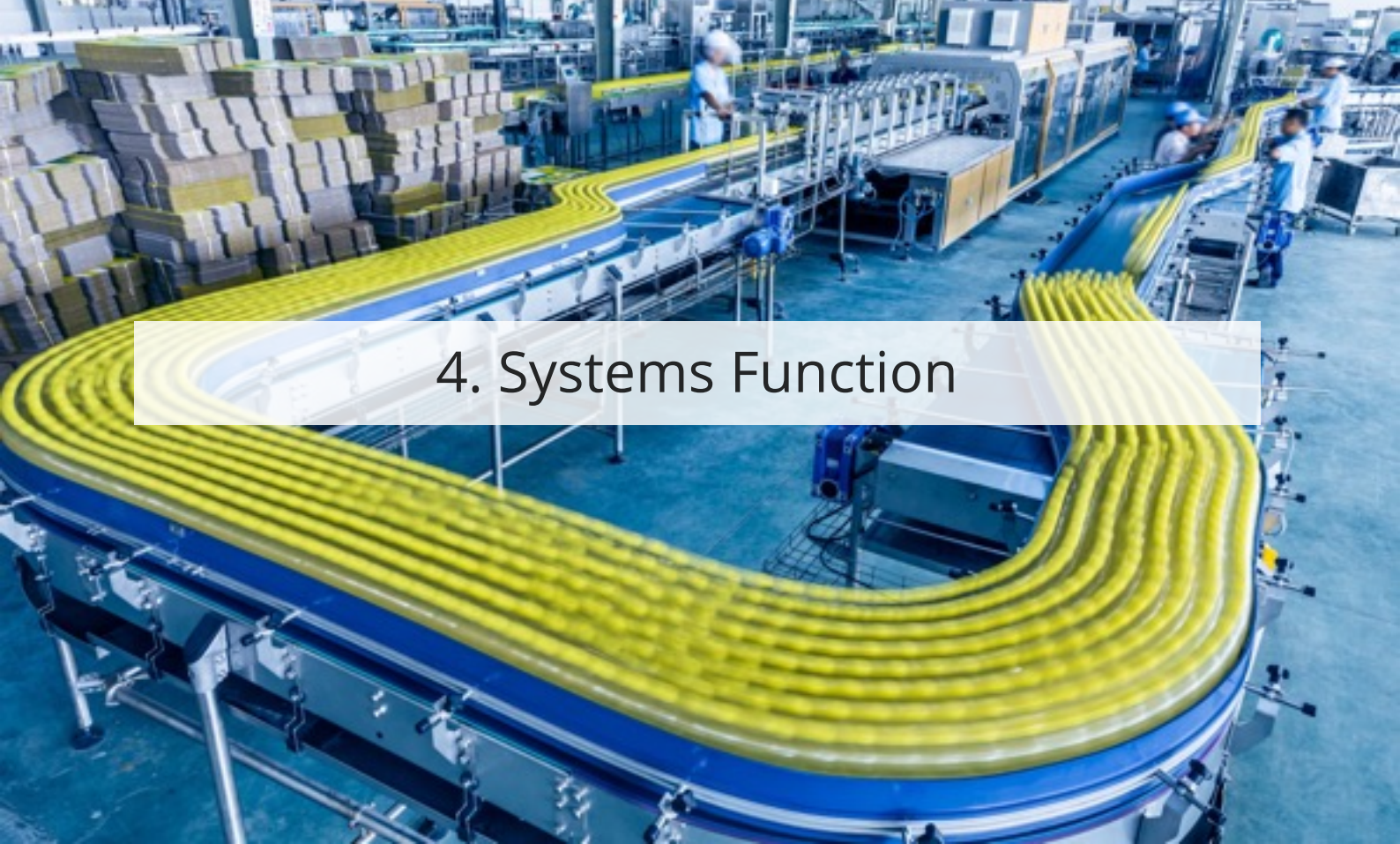


"So, what is a system? A system is a set of things—people, cells, molecules, or whatever—interconnected in such a way that they produce their own pattern of behavior over time." — Donella H. Meadows

This helps to illustration one of the key features to systems and systems thinking, that is what we call emergence which we will be discussing in greater depth later on in the book. But for the moment we will note that a system is not a thing, in contrary to the elements within a system that are things, like bricks, cars, people, planets etc. a system is what emerges out of the interactions of these things when they work together as an entirety. This makes systems and systems thinking a little bit more abstract because we can't really touch, grasp or hold a system. Think of a urban transportation system, we might be able to see a bus or walk on a road but it is difficult for us to grasp the whole that is the system.

Where as the elements within a system have much more well defined boundaries the system as a whole is a much more open and nebulous thing. Thus it is not surprising that we often resort to using analytical methods whereby we simply describe the system by describing all of its parts, thus reducing it to a simple set.

Now that we have an overview to analysis and syntheses from the previous section and know what sets and systems are from this section we should be able to see how the two relate. When we are dealing with sets we use what is called set theory, set theory is essentially the foundations of contemporary mathematics and thus by extension contemporary science, which both represents the analytical method of reasoning. And not surprisingly when we are dealing with systems (such as social systems, transportation systems, ecosystems, information systems etc.) we need to use systems thinking that is based upon synthesis.



4. Systems Function

As we have previously discussed systems are essentially the global functionality that emerges out of the interaction and arrangement of a set of elements. Thus systems are defined by the function that they perform, to see the world from the systems perspective is to see not things but to see their functions. So in this section we are going to talk about these things called functions; a function is a very broad and fundamental concept that is central to systems theory, it is also used in many different domains being particularly important within mathematics and engineering.

Put simply a function is a process that transforms energy or resources from one state to another. So there are three key things to note in this definition, we have a set of things that are the input, we have a process that changes these things in some way and we have an output. Firstly inputs involve the capturing and assembling of elements that enter the system to be processed, putting fuel into your car is an example of an input to a mechanical system, the water taken in by the roots of a plant is an example of an input to a biological system. An important thing to note is that any given system can only process a specific range of inputs, our car can process a certain type of fuel but not all fuels. As we will discuss in a later section what a system can and can't process is a defining feature to its boundary that functions to filter inputs to the system. For example an electrical power socket is designed in a particular shape to ensure that only the right plug is inputted to it thus it functioning as a boundary filter to accessing the system.

Secondly resources that are successfully inputted are processed within the system. A process is a series of actions performed upon the input in order to achieve a particular end result. Processing is often understood in terms of information, that is to say an algorithm or set of instructions that are performed upon the input in order to produce the output. So baking a cake is an example of a process, it takes in a set of ingredients, such as eggs, water, flour etc. The cook then has a recipe to follow that represents the set of instructions to be performed upon these raw ingredients, such as chopping, mixing, baking and so on. If these stages in the process are correctly performed the result should be the desired output.

This same process is true for the internal workings of a computer, biological cell or a financial transaction. Processes are not necessarily linear in nature they may be cyclical, feeding back on each other with the output from one process being the input for another, they may also be nested within larger processes or run parallel to them.

But ultimately there will be some energy or resources produced by this process that travels across the system boundary to be returned to its environment and this is what is called the system's output. One way of understanding a function then is simply as the difference between what goes into the system and what comes out.

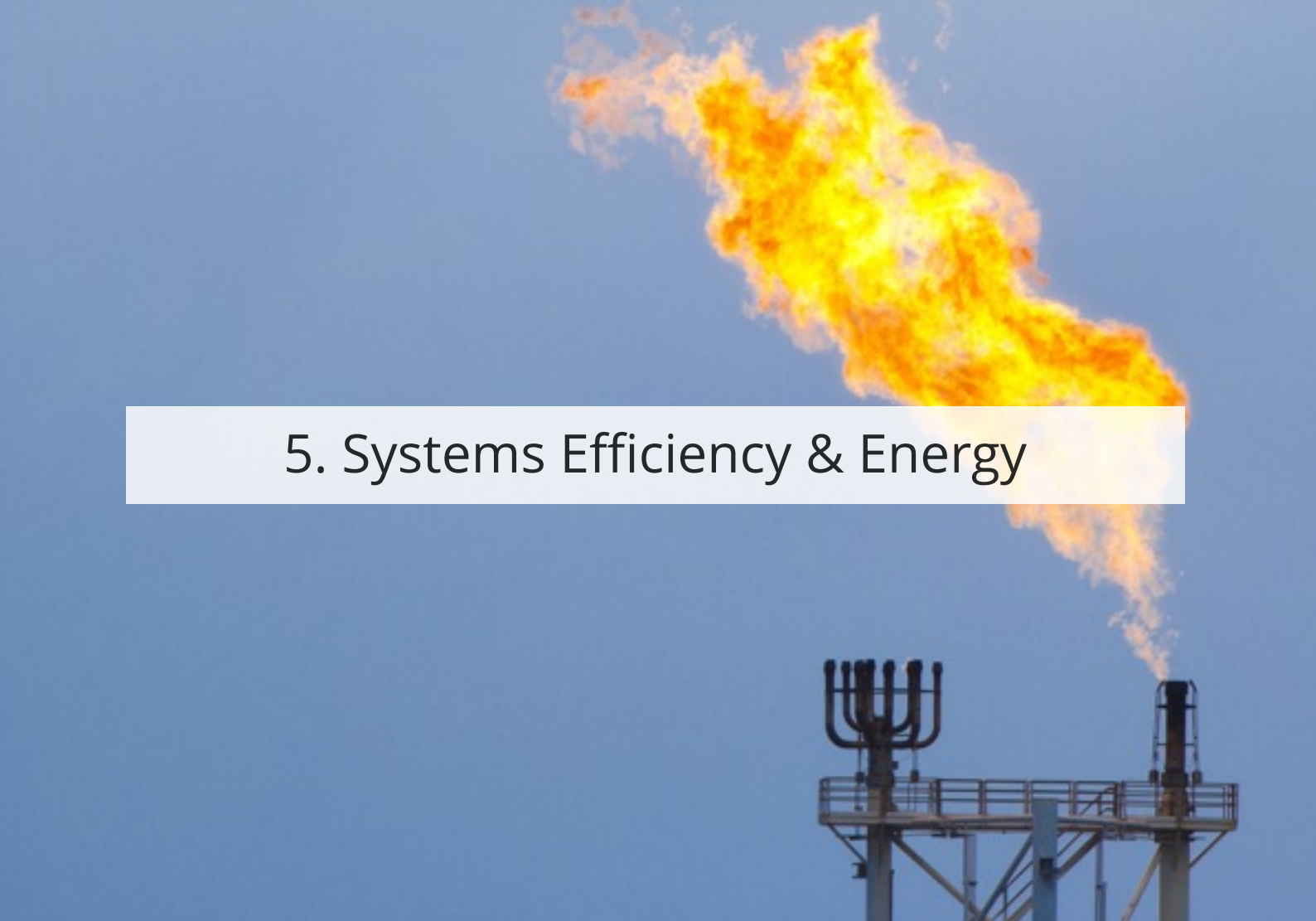


"To enable a system to perform effectively we must understand it—we must be able to explain its behavior—and this requires being aware of its functions in the larger systems of which it is a part"
- Russell L. Ackoff

We call a system whose internal functioning we do not know a black box. In science, computing, and engineering, a black box is a device, system or object which can be viewed in terms of its inputs and outputs, without any knowledge of its internal workings. This can be of great value to us as it helps to hide away the complexity of the internal workings to the system.

A function is often symbolically denoted as an arrow from one element or set of elements to another and in the language of mathematics may be called a mapping or a transformation. Functions may be unidirectional meaning the function only maps an input to an output or the function may be bidirectional meaning it can also invert this process to transform the output back to the input through what is called an inverse function. Many processes are essentially unidirectional requiring vastly or even infinitely more energy to invert the function than was required to perform it in the first place, aging within the human body may be cited as a good example of a unidirectional process whereas the building of a Lego brick house is an example of a function that can be easily inverted and thus bidirectional.

We should note that the model of a function can not be properly used to describe a set of things, because sets do not perform a common function. So if we take a group of nations at war with each other, because they are not working together to perform a function we can only describe them by talking about their attributes and interactions. But the model of a function can be effectively used to describe any type of system where the components are working together towards some common end. The concept of a function will appear to be very simple and intuitive to us this is due to its high level of abstraction, which also makes it a powerful model and very important tool in our systems thinking toolbox.



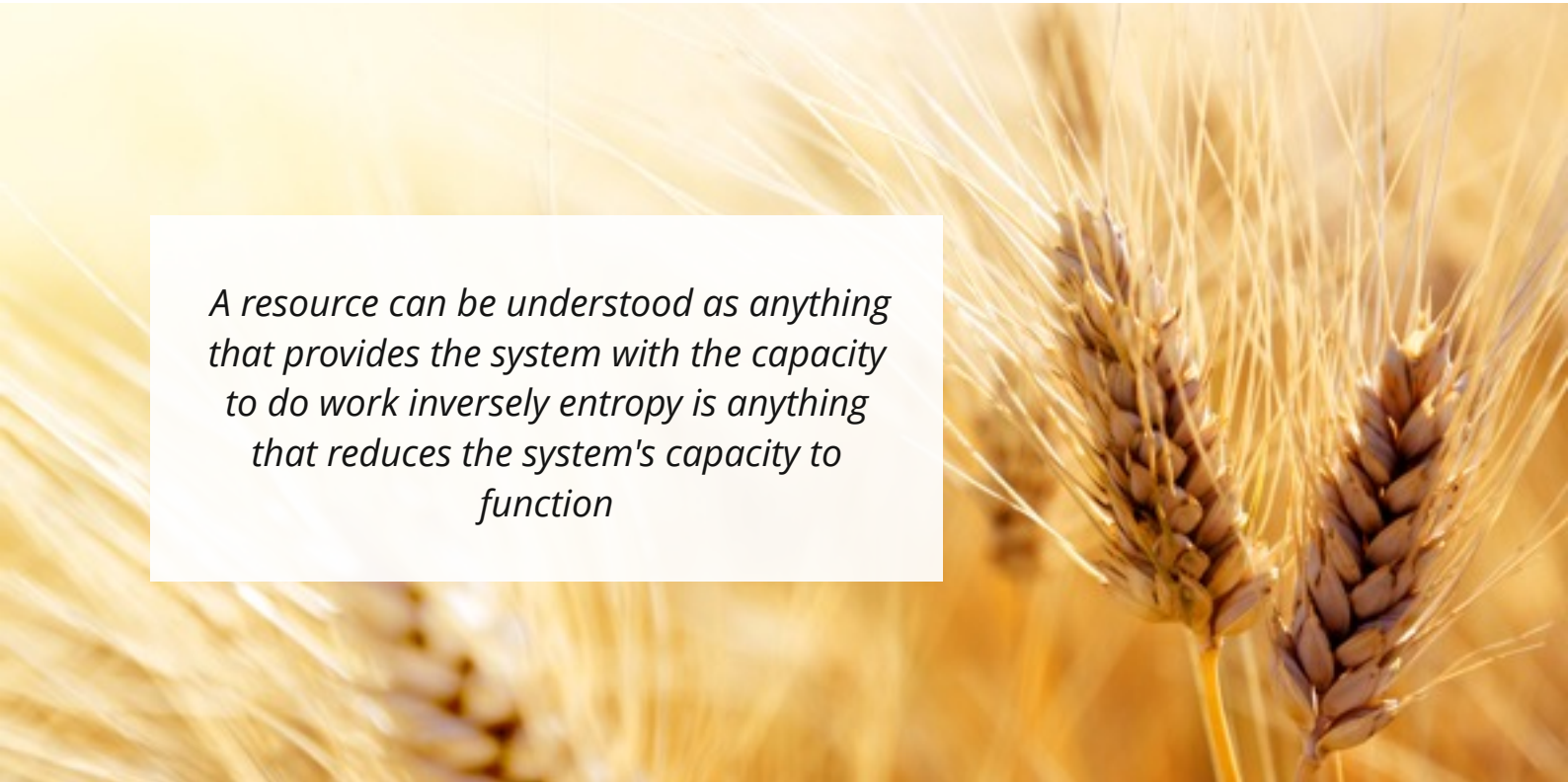
5. Systems Efficiency & Energy

In the previous section we talked about systems functioning but purposefully left out some important aspect to the process. That is to say what the system is processing and how well it does this. Whether we are talking about a car, political system or a farm we are often interested in answering the question how well does it work. That is to say what is the ratio between the resources that the system takes in and those that it outputs, in the language of systems theory this is called the systems efficiency. Answering this question may not be too difficult if we are simply talking about something like a steam engine. But if we wish to be able to reason about all types of systems in this way the question becomes a little bit more difficult than it might sound, so we need to start by being clear about some of the terms we are using.

Firstly a resource is essentially a stored form of energy, an ordered structure that enables a system to perform work. Examples of this might be the food that humans metabolize in order to fuel our bodies or petroleum where energy is stored in chemical bonds. The opposite of energy is entropy, which is the incapacity to perform work and a measurement of the degree of disorder within a system. Whereas a stored form of energy is called a resource a stored form of entropy may be loosely equated to the term waste. An example of an entropic system might be a vase that has fallen

on the ground and shattered, the parts are arranged in a random unordered fashion making them incapable of serving their intended function.

Energy and entropy are typically measured using information theory, that is to say we can measure the degree of order or disorder within a system in terms of the information correlation between its constituent elements, the more patterns there are between the parts the less information it will take to describe the system and thus the more ordered it is said to be. Thermodynamics is the area that studies energy in relation to heat; whilst energetics is the area that studies energy on a broader level within all physical systems and it is closely related to systems theory.



A resource can be understood as anything that provides the system with the capacity to do work inversely entropy is anything that reduces the system's capacity to function

The functioning of a system can be either productive or consumptive. A function can be said to be productive if the system takes in some resource from its environment and performs work on this resource by transferring energy to it and thus outputting a resource of greater value. An example of this might be simply lifting an object off the ground, when we model this phenomena as a system we can see that we inputted an object at a low level of potential gravity and in transferring energy from our selves to the object (by lifting it) we outputted an object at a higher state of potential energy. A resource that now has a greater capacity to perform work than it did before we performed this operation on it.

Inversely a function can be said to be consumptive when the resource that was inputted transfers its energy to the system, conserving this energy within the system's boundary whilst outputting entropy (other wise known as waste) to its environment. Probably the simplest example of this is the metabolic process of digestion within mammals here the resource of food is inputted to the system, energy is extracted from it and a waste product of excretion is exported from the system.

So we can define systems efficiency as a ratio between energy inputted and the energy outputted. But unfortunately what is considered energy and what is considered entropy is by no means objective and is often relative to the system's environment. As an example of this we might think about a lightbulb consuming some amount of electricity as its input and produces some amount of light as its output. Though not all the electricity is converted to light, some is converted to heat energy, with respect to the functioning of the lightbulb as a light producer this heat would be considered waste or entropy. But if we were interested in heating our house then this excess heat energy may be considered a resource. In order to understand this better we need to think outside of the systems, that is to say start talking about the environment and this is where we will pick up in the next section.



6. Boundaries & Environment

Up until now we have been talking primarily about the internal workings of systems, but in this section we will start to present models for understanding systems within the context of the broader environment that they operate in. The first thing we need to discuss is what is called the system boundary. The system boundary demarcates a limit to the system's internal components and processes. Internal to its boundary the system has some degree of integrity, meaning the parts are working together and this integrity gives the system a degree of autonomy. But that is all quite abstract so let's take some examples.

If we take a tree for example every part of the tree has been designed in some way to function as part of the entire system, the bark, leaves and trunk all serve some function with respect to the whole and thus they are integrated and through this integration, they are able to function independently from other systems in their environment. Thus the leaves in a tree are dependent upon the tree's trunk and all the other elements of that tree but they are independent from the leaves and trunks of other trees, that is to say the tree as an entirety has a degree of autonomy.

A system's boundary is then demarcated by where the nexus of relations that enable it to function as an integrated and autonomous whole reach their limit, beyond this the system loses its autonomy and has to interact with other systems and its environment. The boundary to a nation state is another example of this, the nation's border is only a boundary if within this boundary public functions are integrated within the national system as an entirety and by the nation functioning as a whole it can be autonomous from other nations. If one region of this nation has a different culture from that of its parent nation, instead sharing its heritage with a neighboring

country, this will reduce the internal integrity of the nation, its autonomy to act as an entirety and reduce the degree of definition to its boundary. These examples should hopefully help to illustrate that boundaries may have a physical dimension but can't always be defined in physical terms. If we want to be able to achieve sufficient generality to talk about all types of systems, which we should remember is the aim of systems theory, then we need to understand boundaries within this slightly more abstract language of integrity and autonomy.

Within the language of systems theory, systems are said to be open or closed. Open systems interface and interact with their environment, by receiving inputs and delivering outputs external to their boundary. These boundaries are permeable meaning that they may permit the exchange of materials, energy, information or ideas. Inversely, closed systems are more prone to resist incorporating new inputs and this resistance at their boundary makes them more strongly defined by the static properties of the boundary. By not adopting inputs, a closed system ceases to properly serve a function within its environment, may become deemed unnecessary to its parent environment and risks atrophy.



"A system is closed if no material enters or leaves it; it is open if there is import and export and, therefore, change of the components. Living systems are open systems, maintaining themselves in exchange of materials with environment, and in continuous building up and breaking down of their components" - Von Bertalanffy

An isolated system is more restrictive than a closed system as it does not interact with its surroundings in any way. The universe as an entirety might be an example of an isolated system, but it is debatable as to whether such a construct could exist in reality. So let's take a few quick examples of open and closed systems to try and cement the idea. A hospital is an example of an open system continuously taking in new patients and discharging others, receiving medical requirement and removing old, hiring new personnel whilst retiring others. This high rate of input and output to open systems make them dynamic, they are constantly changing and have to respond to the changes within their environment. An example of a closed system might be a boat on the sea it is specifically designed not to take in water from the oceanic environment it is a part of. Another example might be a group of teenage friends in a public park, engrossed within the internal cultural dynamics of their peer group they are capable of receiving only a very limited input of impressions from their broader environment.

Finally we get to the system's environment; all systems have a boundary and operate within an environment. This environment represents the sum total of other systems and input/output resources that the system interacts with during its operation. Thus the environment consists of the sum total of resources and systems that lie outside of the boundary of the system of interest and interact with it providing its inputs and receive its outputs. From this we should note that a system's environment is primarily relative to its functioning. So a biological system that requires the input and output of natural resources operates within the natural environment. A business or enterprise system that requires the input of economic resources operates within a given market environment. And the political system of a nation operates within the international political environment. We will wrap-up here and continue our discussion in the next section where we will be talking about the relations between elements and systems.

A close-up photograph of several hands assembling interlocking puzzle pieces. The pieces are in various colors: red, orange, white, and green. The hands are positioned around the pieces, with some fingers holding them steady and others placing new pieces into the assembly. The background is a soft, out-of-focus blue.

7. Synergies and Relations

As the famous scientist Carl Sagan once said; “The beauty of a living thing is not the atoms that go into it, but the way those atoms are put together.” This short quote goes to the heart of the systems paradigm and tells us why it is the relationship between components that we are really interested in when seeing the world from the systems perspective. A relation is a simple but abstract concept; it is a connection or interaction between two or more components. Through this connection there is an exchange of some matter, energy, information or ideas that bind the elements into a state of interdependency, where the total gains and losses of any component are correlated with those of others in the relationship.

These relationships between the systems constituent elements can be fundamentally of two different kinds, constructive or destructive. We call constructive relations synergies and destructive relations interference. Starting with synergies. A synergy is an interaction or cooperation of two or more components to produce a combined effect greater than the sum of their separate effects.

A classical example of a synergy is the relationship between the honeybee and the flowers it pollinates. Bee and plant interact by exchanging pollen and nectar, both elements have a need that they can not fulfill themselves. The bee needs some resource for its subsistence which it can not produce itself and the plant that is incapable of mobility needs some form of transportation for its pollen. It can be said that they both get out of this interaction more than they put in and thus the sum total is greater than the simple combination of their resources in isolation. Examples of synergistic phenomena are ubiquitous in the natural world but another example of a synergy could be the increased gains resulting from a business merger, which can be attributed to various factors, such as combined talent or economics of scale and cost

reduction. With synergies the value added by the system as a whole, beyond that contributed independently by the parts, is created primarily by the relationship among the parts, that is, how they are interconnected.



Synergy means two or more things working together in a constructive fashion to create a combined output that is greater than the sum of the parts

In contrast to synergistic relations we also have relations of interference that are destructive in nature, meaning they reduce the combined output of the system to less than the sum of its parts. Interference is the prevention of a process or activity from being carried out properly due to some interaction between elements or systems. An example of this would be the interference between two drugs, a situation in which a substance affects the activity of another drug in negative ways when both are administered together. Thus reducing the overall positive effect to less than the benefit of the individual effects in isolation. Another example of this is destructive interference between sound waves, where sound waves that are out of sync lead to their canceling each other out. Thus we can see how the degree of synchronization or a-synchronization between elements is an important factor in determining the nature of the relations between them.

We can also note that synergies often arise as a product of differentiation and specialization. Differentiation is a process that occurs in many systems as they develop, it is defined by the proliferation of subsystems and specialized elements internal to the system in order to make it more capable of responding to a greater

diversity of states within its environment. Differentiation occurs most notably during the development of a multicellular organism, which originates as a single cell but through cellular division the organism develops a multitude of differentiated, or specialized, cells capable of performing many different functions. The same can be observed in the development of technologies and social organization.

The point to take away from this is that this process of differentiation also involves the proliferation of relations with which the now specialized components can avail of each other's services. For example as the global economy has grown with different areas focusing on their specialized domains we have also seen the proliferation of trade relations and this process of specialization and then exchange is a key source of synergistic relations. In the next section we will be carrying on our discuss of synergies when we talk about the closely related topic of emergence, which many would argue is the central idea within systems theory.



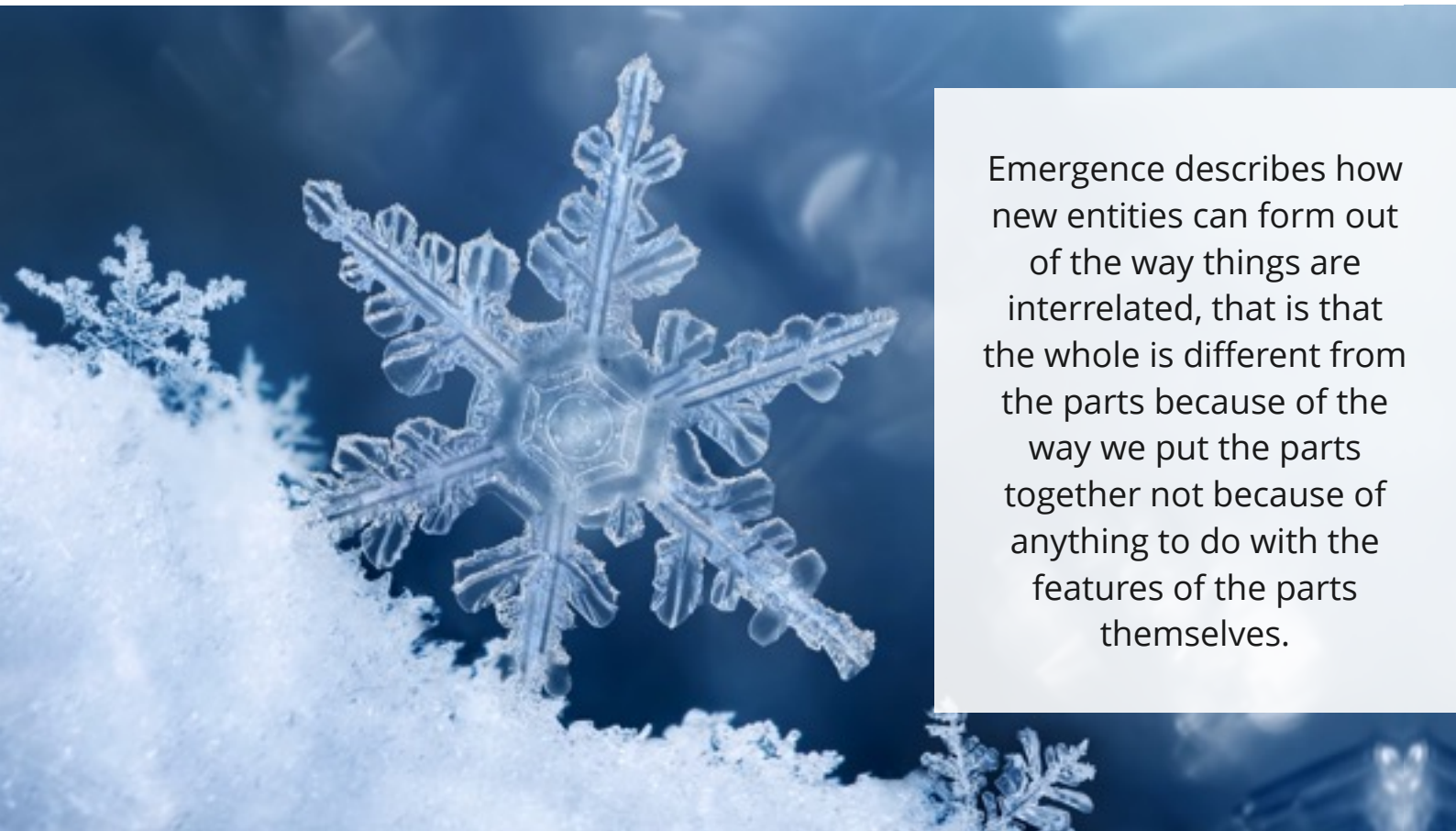
8. Emergence

According to Wikipedia emergence is conceived as - "a process whereby larger entities, patterns, and regularities arise through interactions among smaller or simpler entities that themselves do not exhibit such properties." In the previous section we discussed how synergistic relations give rise to the phenomena of two or more elements having a greater combined output or effect than the simple product of each in isolation. This process whereby the interaction between elements gives rise to something that is greater than the sum of their parts is called emergence.

So where as when we were talking about synergies we simply said that the combined effect was greater than its parts in isolation. The concept of emergence though implies that what is created out of these synergic relations is not just quantitatively different it is in fact qualitatively different. That is to say none of the elements that contribute to the emergence of this new phenomena have its qualities when taken in isolation.

There are many examples of this but maybe the simplest is the example of water, water is made up of hydrogen and oxygen atoms, neither of these two elements that make up the system have the property or quality of wetness, but when we combine them we get a substance called water that has the quality of being wet, this property of wetness has emerged out of the interaction of the systems elements and it only exists on the macro level of the whole system. Another often cited example of emergence is the phenomena of life, biological systems such as a plant cell consist of a set of inanimate molecules none of which in isolation have the property of life, but it is the particular way that these elements are arranged into structures and processes that enable the emergent phenomena of the living system as an entirety.

Our world is full of examples of emergence that we could site, from ant colonies to galaxies and cultures, but all of these are types of structures, where as emergence is really a process. These systems are then the product of the process of emergence that has played out to create two qualitatively different levels to the system. Emergence then is a process through which systems develop or we might say grow. During this process unassociated elements interact, synchronize to form synergies and out of this emerges some new and novel phenomena that previously did not exist.



Emergence describes how new entities can form out of the way things are interrelated, that is that the whole is different from the parts because of the way we put the parts together not because of anything to do with the features of the parts themselves.

In order to create some qualitatively different and new phenomena the system must go through what we call a phase transition. A phase transition is an often rapid or accelerated period during the process of a system's development, either side of which the fundamental parameters with which we describe the system change qualitatively.

Again there are lots of examples of this such a the phase transition between solid and liquid that a substance goes through when heated, but maybe the most dramatic example is the metamorphose of a butterfly from being a caterpillar to a mature adult. Not only dose the system's morphology change but the whole set of parameters that we define it with are so drastically altered prior and post the phase transition that we give the creature a whole new name.

This illustration helps to bring us to another important theme within emergence, that is the distinction made between what is called strong and weak emergence. Weak emergence describes how the emergent phenomena can be traced back to the individual elements. Meaning we can predict and observe higher-level emergent phenomena just by looking at individual components. In contrast, strong emergence, also known as "irreducible" emergence, states that these phenomena cannot be reduced to the individual components. Instead, the emergent phenomena are traced back to the interactions between the multiple components, so quite literally *cannot* be predicted in any sense by looking at the components on their own. Consciousness is often cited as an example of strong emergence, it would appear that without prior knowledge or experience of what consciousness is, it would be virtually impossible to understand the vastly complex and subtle system that is human consciousness by analyzing the properties of the very simple neurons that formulate it.

This distinction between strong and weak emergence may also be formulated within the language of information theory. Where weakly emergent phenomena are essentially computable, that is to say if we have sufficient information we could simulate them, in contrast with strongly emergent phenomena where no amount of information could predict or formulate the end result of the process prior to its completion. The discussion of strong and weak emergence leads us to another key theme in systems theory that is hierarchy, the distinction between micro and macro and top down vs. bottom up causality all of which we will be talking about in the next section.



9. Hierarchy & Abstraction

Up until now we have been talking about systems on one level of analysis. Our model so far has consisted of simple elements making up systems, the reality of the world we live in though is of course vastly more complex than this and one way of capturing and structuring this complexity is through the use of abstraction and hierarchical structure. Abstraction is the process of successively removing layers of detail from our representation in order to capture the most essential features to a system. An architect's master plan of a building is an example of an abstract representation, it is designed to capture only the most essential features to the building that are required to get an idea of its overall make up.

By using abstraction we can define different levels to our model depending on its degree of detail or granularity and this is called encapsulation. We are encapsulating one model of a system inside of another which in turn may be encapsulated within a third and so on creating a hierarchical structure within our representation. You might ask what the value of this is? The value of this is that almost all phenomena exhibit this hierarchical structure, whether we are talking about physical systems where atoms make up molecules, which make up substances and so on, or social institutions where individuals make up organizations which make up societies and etc.

In order to give some terminology to these different levels we have at least four different terms we can use, at the most basic level of the hierarchy is what are called elements. Elements are elemental, meaning they do not have constituent components we treat them as a whole, they simply have properties, an electron is an example of an element, we cannot look inside of it because it is not made of any separate parts.

Next up are subsystems, A subsystem is a set of elements, which make up a system, which in turn is a component of a larger system. An example of a sub system might be the breaks in a car, they are made up of elements but are also an integral part of a broader system, the car. Our car, which is a system of personal mobility, is in turn part of a transportation system and we call this level to our analysis a system-of-systems. Lastly all of this is encapsulated within our ultimate unit of analysis, that is the system's environment.

Different types of systems base their hierarchy upon different features, so hierarchies within ecosystems are based upon where creatures lay in the food chain, within social systems hierarchies maybe based upon age, occupation, education or many other factors. The theory of integrative levels tries to describe the underlining dynamics and characteristics of this ubiquitous feature of organizational levels.



The theory of Integrative level deals with the idea that units of matter are organized and integrated into levels of increasing integration and complexity. The idea of integrative levels of organization allow us to describe the evolution from the inanimate to the animate and the social world.

Higher integrative levels are thought to be more complex and demonstrate more variation and characteristics than lower integrative levels. Because of emergence each level has its own unique internal dynamics and can not be fully reduced to the level bellow and thus we have the domains of biology, sociology and cultural studies because novel features to systems emerge on each of these particular levels of

integration, that can not be describe by simple reference to physical structures and process.

The last thing to note in this section is that as soon as we have emergence and hierarchical structure we have a new dynamic between the different levels to the system. As emergence implies that the rules governing any given level may be qualitatively different from those of another and this will be particular pronounced when we take the two extremes of the system's micro and macro level. As all of these different levels have to ultimately work together as an entire system. The question turns to whether it is the rules that govern the micro level to the system or the rules that govern the macro level that ultimately determine the system's functioning as a whole, you may also hear this dynamic referred to as bottom-up verse top-down causality and it is another key theme within systems theory.

So lets take an example of this; If a doctor has a patient that is in poor physical health and psychologically depressed, does she search for a bottom up cause to the systems dis-functionality, in which case she would look for a physiological explanation, something like a virus or infection that is causing the overall problems within the patient's body. Or inversely does she search for a top down explanation reasoning that it is the patience's psychological state that is inducing their physiological state of poor health.

Debating this question farther is beyond the scope of this introduction, but the point to take away is that within these emergent hierarchical system, such as the human body, political regimes or ecosystems, there will always be this complex dynamic between the rules that govern the system on the micro level and those that govern it on the macro level. We can wrap up then by saying that abstraction is a powerful method of reasoning, by using encapsulation to nest sub-systems within systems we can create models that capture the emergent hierarchical structures that we see all around us in the world.



10. Systems Dynamics

Up until now our model of a system has been relatively static. In this module we are going to start to deal with how systems change over time, what is called system dynamics. System dynamics is a branch of systems theory that tries to model and understand the dynamic behavior of complex systems. It deals with internal feedback loops and time delays that affect the behavior of the entire system. It was first developed by Professor Jay Forrester at MIT as a management method but has since go on to be applied to all types of systems from modeling the dynamics of earth's systems to those of the economy and political regimes.

The key elements of system dynamics are feedback loops, stocks and flows. Firstly feedback, with analytical thinking we often see the world in terms of linear cause and effects, but systems thinking looks for the interplay between elements, that is the feedback loops through which elements are interconnected in effecting a joint outcome. System dynamics uses what are called causal loop diagrams to do this. A causal loop diagram is a simple map of a system with all its constituent components and their interactions. By capturing interactions and consequently the feedback loops, a causal loop diagram reveals the structure of a system. By understanding not only the structure to these relations but also the nature of those relations it becomes possible to model and simulate a system's behavior over a certain time period. These feedback loops can then be of two different kinds, either positive or negative.

A Positive feedback loop means that values associated with the two nodes within the relation change in the same direction, so if the node in which the loop starts decreases, the value associated with the other node also decreases. Similarly, if the

node in which the loop starts increases, the other node increases also. Economics of scale is an example of a positive feedback loop between a business and its customers. The more products a company sells the more revenue it receives from its customers giving it more to invest in scaling up production thus allowing it to reduce costs which in turn means more customers will purchase the product and so on. This is also called a virtuous cycle where one party gains the other does so also. Of course this cannot go on forever and that is why positive feedback loops are typically associated with unstable processes that are likely to crash at some time.

A Negative causal link means that the two nodes change in opposite direction, if the node in which the link starts increases, then the other node decreases, and vice versa. The system dynamics between predators and prey are an example of a negative feedback loop. If the number of predators increases then the number of their prey will decrease which will in turn feedback to effect the predators by reducing their pollution, which again will feedback to increase the pray population and so on, negative feedback loops are typically associated with an overall stable and sustainable pattern of development.

There are of course many more examples of positive and negative feedback loops but we will move on to talk about the other key feature to the area of system dynamics that is what we call stock and flow diagrams. To perform a more detailed quantitative analysis, a causal loop diagram is transformed to a stock and flow diagram, which helps in studying and analyzing the system in a quantitative way typically through the use of computer simulations.



"Systems thinkers see the world as a collection of stocks along with the mechanisms for regulating the levels in the stocks by manipulating flows."
- Donella H. Meadows

A stock is the term for any entity that accumulates or depletes over time, thus it is a simple variable. A flow in contrary is the rate of change in a stock. So an example of a stock might be a water reservoir, it is a store of water and we can ascribe a value to the volume it contains. Now if we put a tap on the side of our reservoir and started pouring water out of it, this would be an example of a flow. Where as a stock variable is a measure of some quantity, a flow variable is measured over an interval of time, such as electoral current, which is telling us how fast something is flowing.

By using theses tools of system dynamics we may get a qualitative and/or quantitative idea of how a system of interest is likely to develop over time, for example if we create a simple two dimensional graph with time on the horizontal access, we will see how the different feedback loops create different types of graphs. Graphs for positive feedback loops typical reveal an initial exponential growth as they shoot upwards rapidly but then reach some environmental boundary condition, where they crash back down again. A financial bubble and ensuing crash could be an example of this. Where as the net result of a negative feed back loop will be a wave like graph that will likely be bounded within an upper and lower limit over a prolonged period of time with relatively smooth fluctuations during the systems development that enable it to sustain an overall stable state in the long-term.



11. Homeostasis

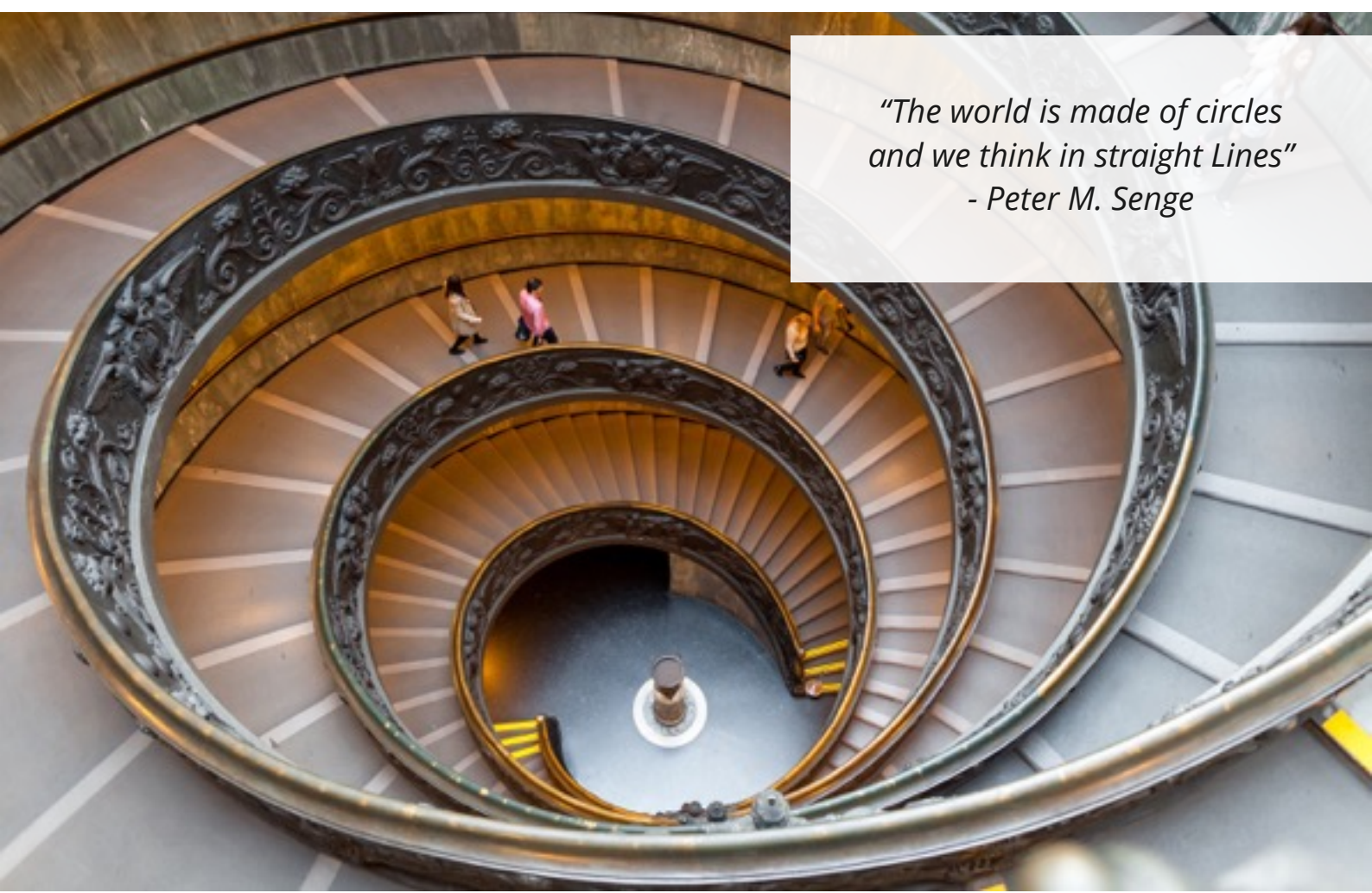
In this section we are going to continue our discussion about dynamic systems within the context of their environment. Many types of systems require both a continuous input of resources from their environment and the capacity to export entropy back to the environment in order to maintain a specific level of functionality. An example of this might be a tractor that must receive a periodic input of fuel and be able to export heat and gases back to its environment for it to maintain its functionality. A business organization is another example, requiring a continuous revenue stream to pay its employees and suppliers while also producing a certain amount of waste material that it must externalize and the same can be said of many other types of systems.

Thus in order for these systems to maintain their intended level of functionality, what we might call their normal or equilibrium state, they must have an environment that is conducive to providing them with these required conditions. Within ecology and biology the term homeostasis is used to describe this phenomenon. The word Homeostasis derives from the Greek word meaning *homos* or "similar" and *stasis* meaning "standing still". It is the state of a system in which variables are regulated so that internal conditions remain stable and relatively constant, despite changes within the system's environment.

In order for systems to maintain homeostasis there needs to be some kind of regulatory mechanism, what we also call a control system, this control mechanism has to regulate both the system's internal and external environment to ensure that the environmental conditions are within the given set of parameters that will enable the

internal processes of the system to function at a normal or equilibrium state. Cybernetics is the area of systems theory that studies these regulatory mechanisms. Cybernetics again comes from a Greek word which means to steer or guide, and this is exactly what a control system is designed to do. It is design to guide the system in the direction of the set of environmental parameters that are best suited for it to maintenance homeostasis, that is to say its maintain its internal composition within just the right parameters required for the needed processes to take place.

So lets take some examples of this; In order to maintain the environmental condition best suited to the physiology of a human being we have invented the thermostat, Thermostats are classical examples of control systems, that operate by switching heaters or air-conditioners on and off in response to the information given by a temperature sensor, thus they regulate the environment in order to maintain a stable or equilibrium condition best suited to the internal workings of the human body. Another example might be the process control system in a chemical plant or oil refinery which maintains fluid levels, pressure, temperature and chemical. There are many more examples of how adaptive systems maintain homeostasis, but the essential characteristic of this phenomenon is to maintain a stable state conducive to perfuming a set of internal dynamic posseses and this is done by monitoring information from feedback loops.

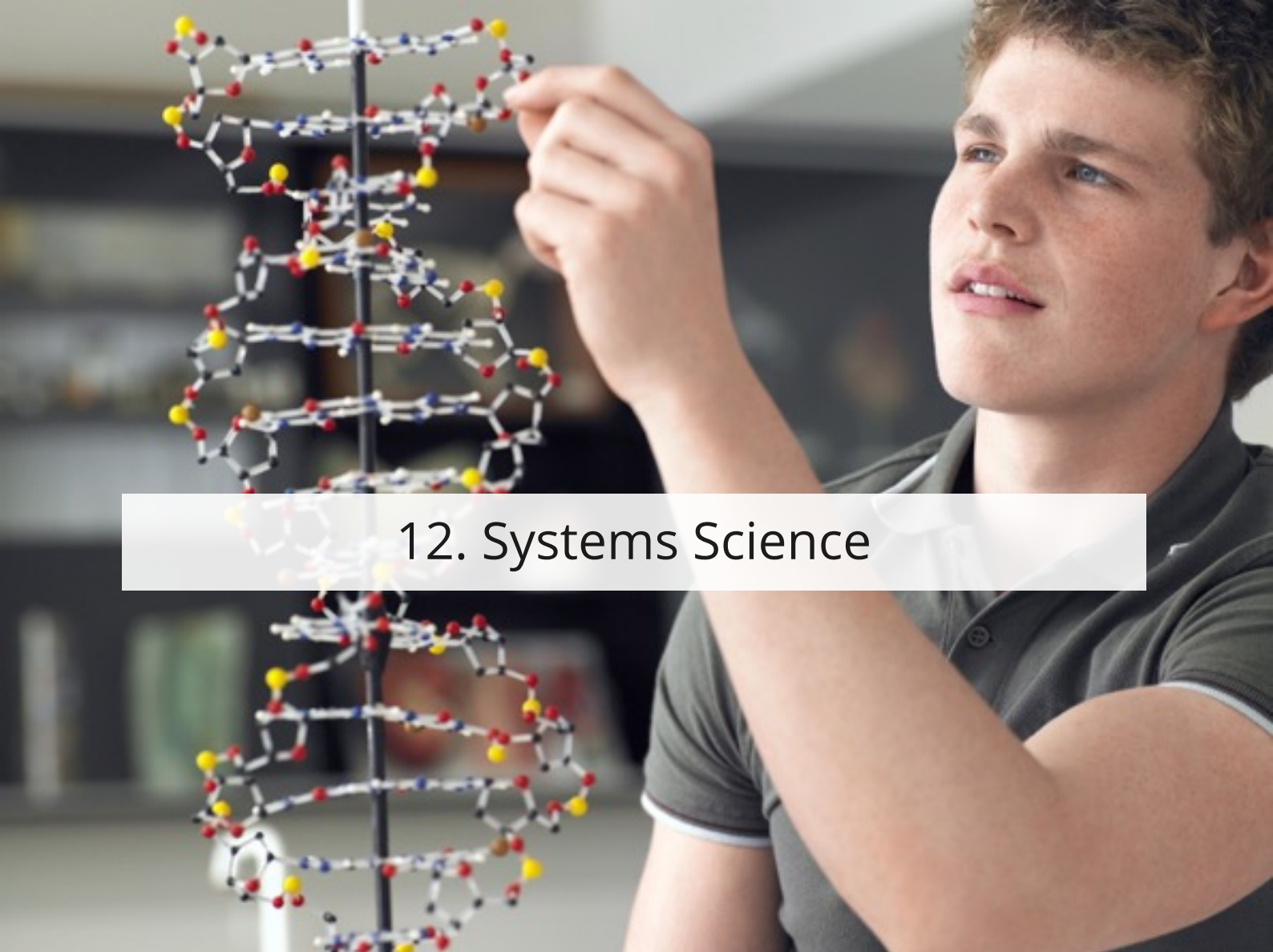


*"The world is made of circles
and we think in straight Lines"
- Peter M. Senge*

If the system is in a homeostatic condition it will simply continue with its previous course of action, but if one or more of the parameters it is designed to monitor are outside of these parameters it will perform some operation in order to effect the state of its environment. The control system then waits for a feedback of information from its environment in order to analyze how this previous activity has adjusted the desired parameters, depending on whether this information signals the system moving away or returning to homeostasis it will again react accordingly.

An example of this might be a person driving a car, when we are cruising nicely along the road we simple continue doing what we have been previously doing whilst also continuing to monitor feedback loops. But as soon as this information signals us approaching the limit of a homeostatic parameter, such as getting too close to the side of the road, we react by adjusting the steering wheel. We then wait a fraction of a second to monitor how this action has effected our status within the environment. Once this information is fed back to us and we have processed it we then once again react all the time with the aim of returning to our desired homeostatic condition that enable the desired function of the car, that is our transit from one location to another.

We can then see how this concept of homeostasis can be a powerful model for capturing the development of any adaptive system, as their course of development is the product of this continuous acting and reacting to feedback loops. Another thing we may note is how complex a system may become give two or more of these adaptive systems acting and reacting to each other's behavior as the system develops through an evolutionary like dynamic. We may also notice how this model captures a lot of the dynamics underpinning the development of social systems such as international politics, free market economies and almost all types of social relations. But this is getting into a whole new area of complex adaptive systems.



12. Systems Science

In this last section to the book we are going to wrap things up by giving an overview to the application of systems theory to the various domains of science, what is called systems science. Systems theory is a formal language meaning like other formal language such as mathematics, it is independent from external reference to any subject matter and thus is solely dependent upon its own internal logic. If this logic is consistent then it works, if there are logical inconsistencies within the syntax of the language then it does not work. The same should be true for any formal language such as the algorithms that run your computer, if there is an error in the programs logic then it will crash the system.

Although the term science in its broadest definition may be used to include the formal languages it is essentially an empirical endeavor, meaning that it is dependent upon reference to some subject matter in order for its validation. Thus the vast majority of people who call themselves scientists spend their time amassing or analyzing empirical data. Whereas the formal languages are independent from empirical science, science works best when it is supported by some formal language and mathematical, as we

know, it is the formal language that supports most of modern science, mathematical proofs are considered the gold standard in terms of scientific validation.

Since the turn of the 20th Century set theory has been the de facto foundations to mainstream mathematics. As we have discussed previously, set theory and the reductionism paradigm are suited to the modeling of certain types of systems. Thus modern science supported by mathematics does a very good job of describing the simple deterministic systems that we have to deal with in the natural sciences, areas like chemistry and particular physics represent powerful, sophisticated and well developed frameworks. But other areas of science, most notably the social sciences, that have to deal with non-deterministic, highly interconnect and emergent systems, either try to mimic the natural sciences, as mainstream economics does, or are left with very little in the way of formal foundations out of which to build any kind of robust framework.

Another aspect to the way in which modern science has developed under the reductionist paradigm is its fractured nature; science is today a highly specialized and compartmentalized activity. Of course there is nothing wrong with specialization, but when the knowledge and expertise of one domain are very disconnect from those of another then science as a body of knowledge can become too focused on the trees without seeing the forest. Science serves a function within society and ultimately a society needs answers to both these analytical questions but also to bigger questions, such the nature of order and chaos in our universe or how the different domains of knowledge really relate to each other.

The reductionist paradigm offers us limited means to approaching these bigger questions and if science cannot provide society with plausible answers then people will look elsewhere and it would have failed in providing us with an integrated picture of how the world works and not just a one sided picture that reduces everything to some simple interaction between physical components.

This is where systems science comes in, with its holistic approach it lets us focus less on specialized knowledge within specific domains and more on how these domains fit together, often through the idea of integrative level. Thus systems science is a much more interdisciplinary form of science being more relevant when we are dealing with phenomena that cross the traditional domains, such as ecology that doesn't confine itself to dealing with biological system but also recognizes the important interplay between human industrial activity, the biosphere and the abiotic geosphere. Thus the area of system ecology has proven one of the most successful areas within the system sciences.

Another interdisciplinary domain systems science is proving particularly relevant to is in the study of the interaction between people and technology, what are called sociotechnical systems. Where as modern science has supported a so what

technocratic view of the world, systems science crosses the two boundaries to recognize the importance of the interaction between people and technology. This leads up to what can possibly be systems science's greatest contribution to our scientific framework. For centuries people have been trying to apply the success of modern physics to studying social systems with limited results. Traditional science rests upon an objective view to the world, that is to say removing the subjective interpretation of the viewer from the model, this works fine when dealing with inanimate objects, but of course there is a subjective dimension to almost everything that humans do. Systems science is philosophically sophisticated enough to deal with the difficult questions surrounding the subjective nature of the human condition, that are require to truly tackle areas like psychology, cultural studies and sociology.

System science and traditional science are often cast in contrary terms but of course they are two sides of the same coin. Developing a scientific framework powerful enough to describe our world in all its richness, will require both the qualitative capacities of systems science, that allow us to properly contextualize things, and the rigorous quantitative methods of analysis that allow us to properly compute this information, with the net result being hopefully a fuller picture of how our extraordinary world works.



Conclusion

This book was designed as an overview to systems thinking, aimed at offering you a brief introduction to the main ideas within system theory in an intuitive and accessible form, hopefully at this state you have an outline to how it works. The most important part of systems thinking we covered in the first couple of sections talking about synthesis and analysis. The central point being that systems thinking is all about shifting our perspective from focusing on the parts and their properties to seeing the connections between things and the whole system. Within this paradigm we are trying to be able to properly contextualize the parts by knowing their function within the whole, how they relate to the other elements within the system and most importantly how all of these interactions give rise to the overall system.

After trying to understand this paradigm that is systems thinking the second major topic we covered was in trying to understand the model of a system. We talked about functions, the efficiency of a function, looking at how processes can be constructive or destructive, consume and generate resources and entropy. We then went on to talk about the relations between elements within a system, introducing you to the idea of interference and how synergistic interactions can give rise to emergence. We talked about a system's bound that defines its autonomy from the environment within which it operates. We looked at the hierarchical structure that is a ubiquitous feature to all kinds of system. Finally we talked about system dynamics, feedback loops and regulation that all help us in understanding how systems change over time.

I hope this book has given you the inspiration to apply systems thinking to your own domain and the interest to pursue the subject farther.